Determinants

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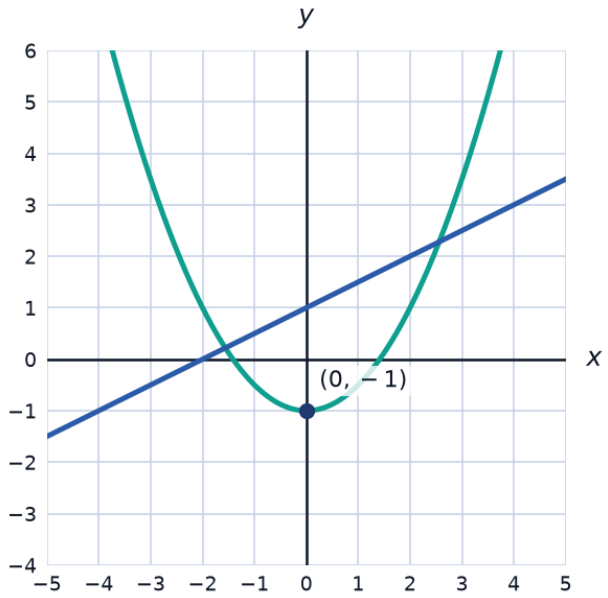
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: 2x2 determinants**

Find the determinant of the matrix with rows $\langle 4, 3\rangle$, $\langle 2, 5\rangle$.

2 **[Foundational] • Target: 2x2 determinants**

Find the determinant of the matrix with rows $\langle -2, 1\rangle$, $\langle 3, 4\rangle$.



3 **[Foundational] • Target: 3x3 determinants via cofactor expansion**

Find the determinant of the matrix with rows $\langle 1, 2, 3\rangle$, $\langle 0, 1, 4\rangle$, $\langle 5, 6, 0\rangle$ by cofactor expansion along the first row.

4 **[Foundational] • Target: 3x3 determinants via cofactor expansion**

Find the determinant of the matrix with rows $\langle 2, 0, 1\rangle$, $\langle 3, 1, 0\rangle$, $\langle 1, 2, 4\rangle$ by cofactor expansion along the first row.

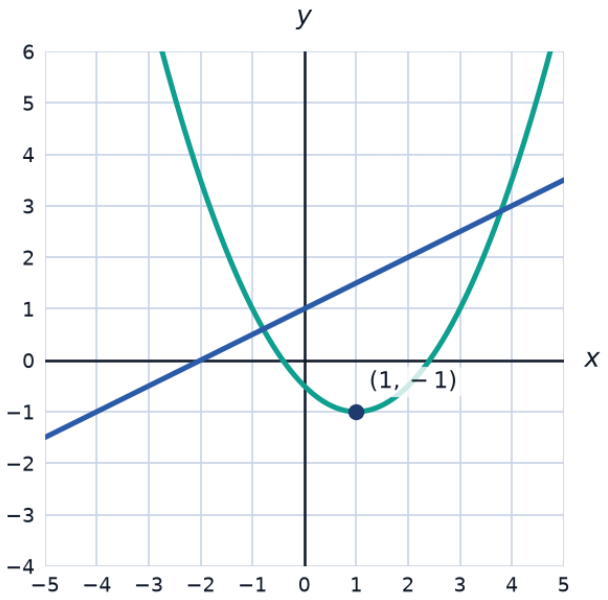
5 **[Foundational] • Target: 3x3 determinants via cofactor expansion**

Recompute the determinant of the matrix with rows $\langle 1, 2, 3\rangle$, $\langle 0, 1, 4\rangle$, $\langle 5, 6, 0\rangle$ by expanding along the SECOND row instead, to confirm the answer matches.

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6 [Foundational] • Target: The determinant product rule

For A with rows $\langle 1, 2\rangle$, $\langle 3, 4\rangle$ and B with rows $\langle 2, 1\rangle$, $\langle 0, 3\rangle$, verify $\det(AB) = \det(A)\det(B)$.



7 [Foundational] • Target: The determinant product rule

Using the product rule $\det(AB) = \det(A)\det(B)$, explain why $\det(A^{-1}) = \frac{1}{\det A}$ for an invertible matrix A.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Determinants and scalar multiples / row operations

If A is a 2×2 matrix with $\det A = 5$, find $\det(2A)$ (scaling EVERY entry of A by 2).

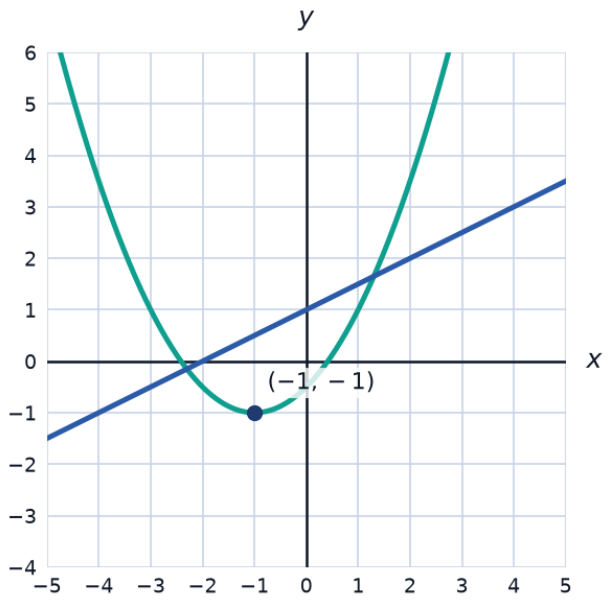
9 [Intermediate] • Target: Determinants and scalar multiples / row operations

If A is a 3×3 matrix with $\det A = 5$, find $\det(2A)$.

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10 [Intermediate] • Target: Determinants and scalar multiples / row operations

Swapping two rows of a matrix negates its determinant. If A has rows $\langle 1, 2\rangle$, $\langle 3, 4\rangle$ with $\det A = -2$, find the determinant of the matrix with rows swapped: $\langle 3, 4\rangle$, $\langle 1, 2\rangle$.



11 [Intermediate] • Target: The invertibility criterion

State the invertibility criterion in terms of the determinant.

12 [Intermediate] • Target: The invertibility criterion

Is the matrix with rows $\langle 2, 4\rangle$, $\langle 1, 2\rangle$ invertible? Justify using the determinant.

13 [Intermediate] • Target: The invertibility criterion

Is the matrix with rows $\langle 1, 0, 2\rangle$, $\langle 0, 3, 1\rangle$, $\langle 2, 0, 4\rangle$ invertible? Justify using the determinant.

14 [Intermediate] • Target: Applying the invertibility criterion

For what value(s) of k is the matrix with rows $\langle k, 2\rangle$, $\langle 3, k\rangle$ NOT invertible?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

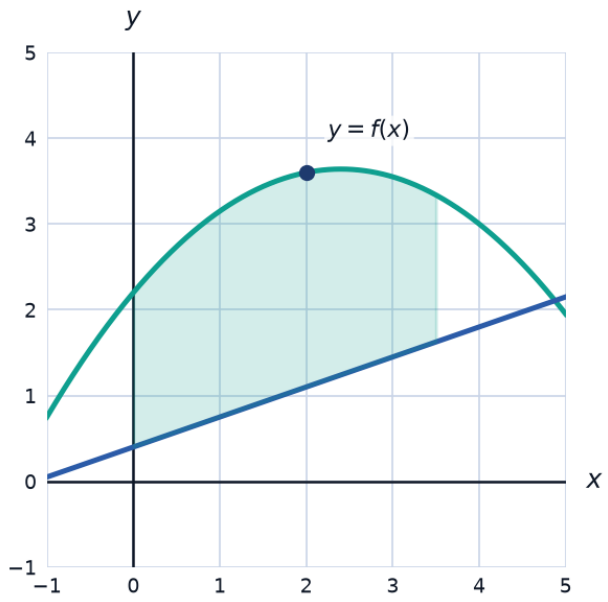
15 [Challenging] • Target: Applying the invertibility criterion • Cross-Lesson: 3.01 Matrices & Matrix Algebra

For what value(s) of k is the matrix with rows $\langle 1, k\rangle$, $\langle k, 4\rangle$ NOT invertible?

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16 [Challenging] • Target: Geometric meaning of the determinant • Cross-Lesson: 1.01 Vectors in n Dimensions

For a 2×2 matrix A with rows $\mathbf{r}_1, \mathbf{r}_2$, $|\det A|$ gives the area of the parallelogram spanned by $\mathbf{r}_1$ and $\mathbf{r}_2$. Find this area for $\mathbf{r}_1 = \langle 3, 0 \rangle$, $\mathbf{r}_2 = \langle 1, 4 \rangle$.



17 [Challenging] • Target: Geometric meaning of the determinant • Cross-Lesson: 2.01 Systems of Linear Equations

Explain, using the geometric meaning of the determinant, why a matrix with two identical rows always has determinant 0.

18 [Challenging] • Target: Determinants of special matrices • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Find the determinant of the identity matrix I_3 (rows $\langle 1, 0, 0 \rangle$, $\langle 0, 1, 0 \rangle$, $\langle 0, 0, 1 \rangle$).

19 [Challenging] • Target: Determinants of special matrices • Cross-Lesson: 1.01 Vectors in n Dimensions

Find the determinant of the matrix with rows $\langle 3, 0, 0 \rangle$, $\langle 0, -2, 0 \rangle$, $\langle 0, 0, 5 \rangle$ (a diagonal matrix).

20 [Challenging] • Target: Determinants of special matrices • Cross-Lesson: 2.01 Systems of Linear Equations

Find the determinant of the matrix with rows $\langle 0, 0, 3 \rangle$, $\langle 0, 1, 0 \rangle$, $\langle 3, 0, 0 \rangle$, and explain why it is nonzero even though the main diagonal has two zeros.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Cofactor Expansion

Evaluate the determinant of $A = [[2, 1, 0], [3, -2, 4], [1, 0, 5]]$.

a. -38

b. 38

c. -22

d. 14

22 [Foundational] • Target: Cofactor Expansion

What is the determinant of any upper triangular or lower triangular matrix?

a. The product of the diagonal entries

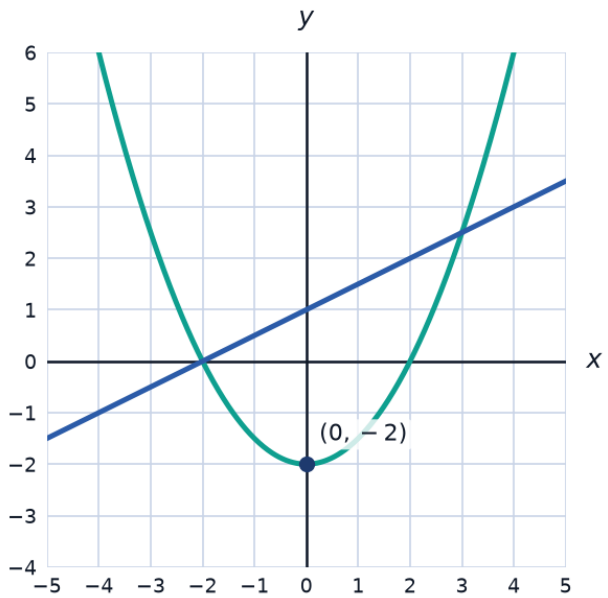
b. The sum of the diagonal entries

c. Always 1

d. Always 0

23 [Foundational] • Target: Cofactor Expansion

If a row of matrix A consists entirely of zeros, what is $\det(A)$?



a. 0

b. 1

c. Undefined

d. -1

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24 [Foundational] • Target: Row Operations and Determinants

How does swapping two rows of a square matrix affect its determinant?

a. The determinant is multiplied by -1

b. The determinant remains unchanged

c. The determinant is squared

d. The determinant becomes 0

25 [Foundational] • Target: Row Operations and Determinants

How does adding a multiple of one row to another row affect the determinant?

a. The determinant does not change

b. The determinant is multiplied by the multiple

c. The determinant is negated

d. The determinant is divided by the multiple

26 [Foundational] • Target: Row Operations and Determinants

If one row of an $n \times n$ matrix is multiplied by scalar c, what happens to $\det(A)$?

a. The determinant is multiplied by c

b. The determinant is multiplied by c^n

c. The determinant is unchanged

d. The determinant is multiplied by c^2

27 [Foundational] • Target: Determinant Properties & Multiplicative Rule

For any $n \times n$ matrices A and B, what is $\det(AB)$?

a. $\det(A) \cdot \det(B)$

b. $\det(A) + \det(B)$

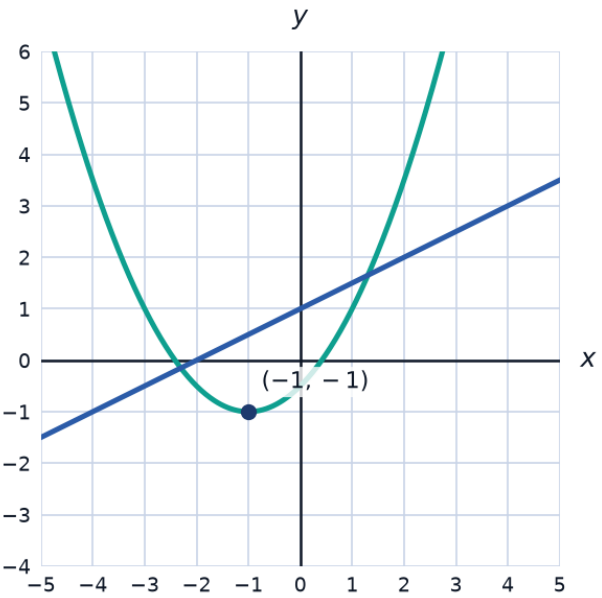
c. $\det(BA) + 1$

d. $(\det A)^{\det B}$

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28 [Foundational] • Target: Determinant Properties & Multiplicative Rule

What is $\det(A^T)$ in relation to $\det(A)$?



a. $\det(A^T) = \det(A)$

b. $\det(A^T) = -\det(A)$

c. $\det(A^T) = 1/\det(A)$

d. $\det(A^T) = (\det A)^2$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Determinant Properties & Multiplicative Rule

Under what condition is $\det(A) = 0$?

a. If and only if A is singular (not invertible)

b. If and only if all entries are zero

c. Whenever A is symmetric

d. Whenever A has positive trace

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30 [Intermediate] • Target: Cramer's Rule and Geometric Interpretations

What geometric quantity does $|\det(A)|$ represent for a 2×2 matrix with columns $\mathbf{a}_1, \mathbf{a}_2$?

a. The area of the parallelogram formed by $\mathbf{a}_1$ and $\mathbf{a}_2$

b. The perimeter of the parallelogram

c. The distance between $\mathbf{a}_1$ and $\mathbf{a}_2$

d. The angle between the columns

31 [Intermediate] • Target: Cramer's Rule and Geometric Interpretations

What does $|\det(A)|$ represent for a 3×3 matrix?

a. The volume of the parallelepiped determined by the column vectors

b. The surface area of the parallelepiped

c. The length of the body diagonal

d. The sum of the lengths of the columns

32 [Intermediate] • Target: Cramer's Rule and Geometric Interpretations

According to Cramer's Rule for solving $A\mathbf{x} = \mathbf{b}$ where $\det(A) \neq 0$, the ith variable is given by:

a. $x_i = \frac{\det(A_i(\mathbf{b}))}{\det(A)}$

b. $x_i = \frac{\det(A)}{\det(A_i(\mathbf{b}))}$

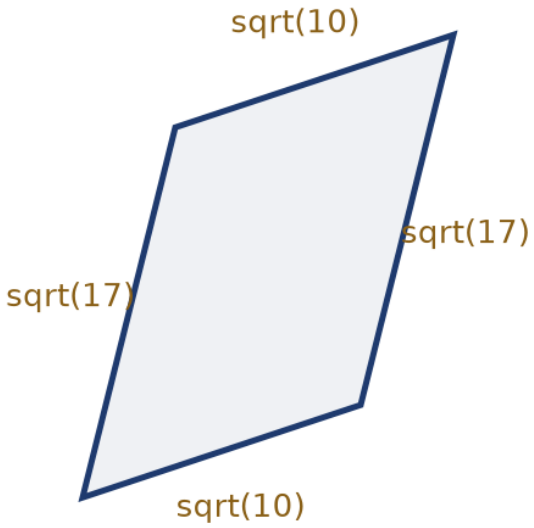
c. $x_i = \det(A) \cdot b_i$

d. $x_i = \frac{b_i}{\det(A)}$

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33 [Intermediate] • Target: Cramer's Rule and Geometric Interpretations

A parallelogram in the xy-plane is defined by the vectors $\mathbf{u} = (3; 1)$ and $\mathbf{v} = (1; 4)$. What is the area of this parallelogram?



a. 10

b. 11

c. 13

d. 14

34 [Intermediate] • Target: 2x2 determinants

Consider the parallelepiped in $\mathbb{R}^3$ defined by the vectors $\mathbf{u} = (2, 1, -1)$, $\mathbf{v} = (1, k, 2)$, and $\mathbf{w} = (3, 4, 1)$. The volume of this parallelepiped is given as 15 cubic units. Find the possible value(s) of k.

a. $\frac{13}{2}$

b. $\frac{15}{2}$

c. $\frac{17}{2}$

d. $\frac{19}{2}$

35 [Intermediate] • Target: 3x3 determinants via cofactor expansion

Consider a matrix $A = (2, -1; 3, k)$. If the determinant of the matrix A^2 is 100, find the positive value of the parameter k.

a. 4

b. 5

c. 2

d. 8

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36 [Intermediate] • Target: The determinant product rule

Consider a parallelepiped in $\mathbb{R}^3$ formed by the origin and the three vectors $\mathbf{u} = (2\ 1\ k)$, $\mathbf{v} = (1\ 3\ 0)$, and $\mathbf{w} = (0\ 2\ 4)$. The volume of this parallelepiped is given as 24 cubic units. Find the value of k if $k > 0$.

a. 2

b. 3

c. 4

d. 6

37 [Intermediate] • Target: Determinants and scalar multiples / row operations

A solid object in $\mathbb{R}^3$ is defined by the region bounded by the planes $x = 0$, $y = 0$, $z = 0$, and the plane passing through the points $P = (6, 0, 0)$, $Q = (0, 4, 0)$, and $R = (0, 0, 3)$. If a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ is represented by the matrix $M = (1, 0, 0; 0, 1, 0; 0, 0, 2)$, what is the volume of the transformed solid object?

a. 12

b. 24

c. 36

d. 48

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The invertibility criterion • Cross-Lesson: 3.01 Matrices & Matrix Algebra

A parallelogram in the xy-plane is defined by the vertices $(0, 0)$, $(3, 1)$, $(1, 4)$, and $(4, 5)$. What is the area of this parallelogram?

a. 11

b. 13

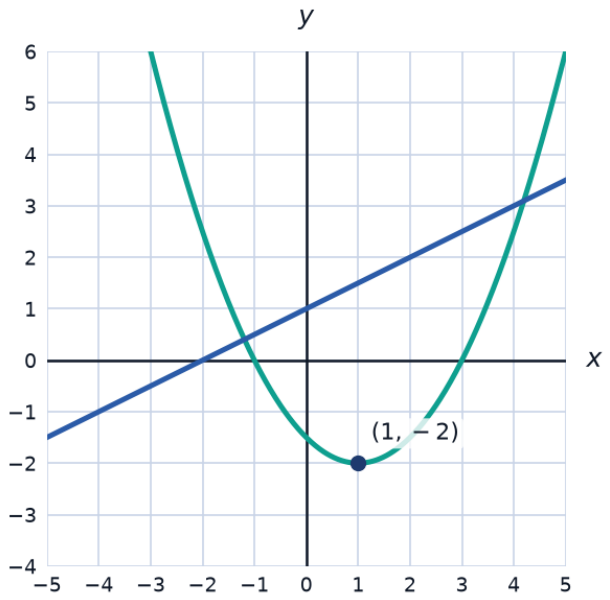
c. 15

d. 17

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39 [Challenging] • Target: Applying the invertibility criterion • Cross-Lesson: 1.01 Vectors in n Dimensions

For $A = [(-1, -3; 2, 3)]$, $B = [(-4, -3; -4, -2)]$, find row 1 of AB.



a. [-2, 3]

b. [9, 16]

c. [4, 6]

d. [16, 9]

40 [Challenging] • Target: Geometric meaning of the determinant • Cross-Lesson: 2.01 Systems of Linear Equations

For $A = [(-1, -1; -3, -1)]$, $B = [(3, 4; 2, -1)]$, find row 2 of AB.

a. [1, -1]

b. [-9, 1]

c. [-11, -11]

d. [-20, -11]

41 [Challenging] • Target: Determinants of special matrices • Cross-Lesson: 3.01 Matrices & Matrix Algebra

For $A = [(-4, -1; -4, -3)]$, $B = [(-3, -4; 2, -3)]$, find row 1 of AB.

a. [28, 15]

b. [19, 10]

c. [12, 3]

d. [10, 19]

42 [Challenging] • Target: Cramer's Rule • Cross-Lesson: 1.01 Vectors in n Dimensions

For $A = [(-3, -1; -2, -1)]$, $B = [(2, -4; -3, -1)]$, find row 1 of AB.

a. [2, 2]

b. [-3, 13]

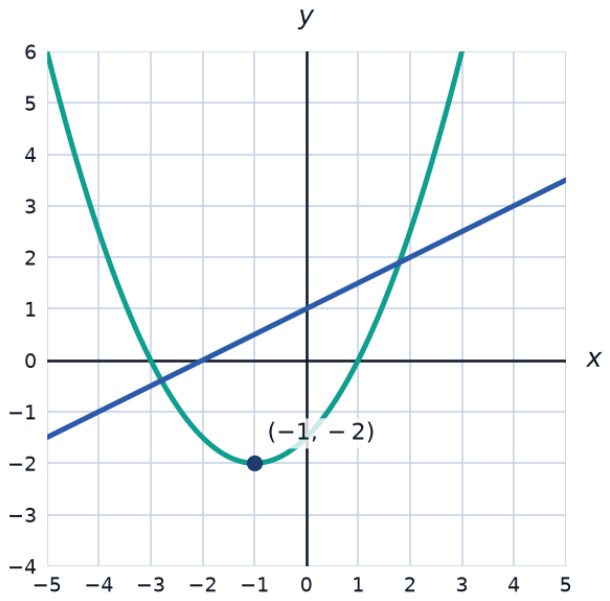
c. [13, -3]

d. [-6, 1]

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43 [Challenging] • Target: Cofactor Expansion • Cross-Lesson: 2.01 Systems of Linear Equations

For $A = [(4, -3; 2, 4)]$, $B = [(-3, -3; 4, -1)]$, find row 1 of AB.



a. [-18, -3]

b. [-9, -24]

c. [-24, -9]

d. [-12, 3]

44 [Challenging] • Target: Row Operations and Determinants • Cross-Lesson: 3.01 Matrices & Matrix Algebra

For $A = [(3, -2; -1, 2)]$, $B = [(-2, -3; -2, -3)]$, find row 1 of AB.

a. [-3, -2]

b. [-2, -3]

c. [-6, 6]

d. [-8, -3]

45 [Challenging] • Target: Determinant Properties & Multiplicative Rule • Cross-Lesson: 1.01 Vectors in n Dimensions

For $A = [(-1, -2; -4, 2)]$, $B = [(-4, -2; -2, -2)]$, find row 1 of AB.

a. [12, 4]

b. [8, 6]

c. [6, 8]

d. [4, 4]

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